

SMART GRID AGENTS FOR EGYPT

EgyptGrid-118

Augmented Lagrangian OPF-RL with Graph Attention Networks & Proximal Policy Optimization

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Environment: Google Colab (CPU / GPU Runtime)

Simulator: Grid2Op (LightSim2Grid C++ backend)

1. PROJECT OVERVIEW

What the System Does

EgyptGrid-118 is a deep reinforcement learning controller that manages real-time power dispatch across a 118-bus electricity network. The agent continuously monitors every bus's voltage, line thermal loading, and generator states, then issues three types of actions every control step: active power redispatch (ΔP), voltage setpoint adjustment (V), and renewable curtailment fraction. It learns entirely through interaction, receiving reward signals that encode both engineering objectives (keep voltage within $\pm 6\%$, avoid line overloads, minimize cost) and equity constraints (fair load distribution across buses).

The system evolves through four phases of increasing sophistication, beginning with a simple reward-based baseline and culminating in a fully augmented Lagrangian formulation with spatially-aware graph neural networks.

Why It Matters

- Egypt's grid is exposed to supply shocks from Suez Canal gas disruptions, renewable intermittency, and aging infrastructure.
- Traditional Optimal Power Flow (OPF) solvers run offline and cannot react to sub-second disturbances or cascading failures in real time.
- A trained RL agent provides millisecond-latency control decisions without solving a non-convex optimization problem at runtime.
- The project grounds its simulation in real 2022 EEHC (Egyptian Electricity Holding Company) generator data, making results operationally relevant — not purely academic.

Real-World Context

Egypt operates one of Africa's largest interconnected grids with over 63 major generating plants including gas, oil, solar, wind, and hydro. Suez Canal transit fees represent a critical vulnerability: geopolitical disruption of LNG/fuel shipping directly threatens the Eastern Canal fossil fleet capacity. This project explicitly models those shocks as a training curriculum, preparing the agent for real operational scenarios.

2. PROBLEM STATEMENT

The Engineering Problem

Optimal Power Flow (OPF) is the core problem of modern grid operation: determine the active power output of every generator and the voltage setpoint at every bus such that total generation cost is minimized while satisfying Kirchhoff's laws, voltage band constraints (typically $\pm 5\text{--}10\%$ of nominal), thermal line ratings, and generator capacity limits. This must be solved continuously — every few seconds in modern EMS/SCADA systems — under load uncertainty and renewable variability.

For a 118-bus, 62-generator system, the AC-OPF is a non-convex, non-linear program with 186 continuous control variables and hundreds of inequality constraints. Adding supply shocks, renewable intermittency, and fairness objectives makes this even harder.

Limitations of Traditional Approaches

Approach	Key Limitation
Interior-Point OPF	Computationally expensive at large scale; assumes full stationarity — cannot handle disturbances mid-solve.
Rule-Based Control	Hand-crafted rules cannot generalize to novel supply shocks or topology changes.
Convex Relaxation (SDP/SOCP)	Relaxation gaps grow under severe loading; may return infeasible operating points.
Standard RL (no constraints)	Agent ignores voltage/thermal limits unless manually encoded — leads to unsafe dispatch.

This project addresses the gap by combining the constraint-handling power of Lagrangian optimization with the generalization capability of deep RL — enabling a controller that is both physics-aware and adaptive.

3. METHODOLOGY

3a. Environment & Data

The simulation runs inside Grid2Op (l2rpn_wcci_2022) backed by LightSim2Grid, a C++ power-flow engine that solves full AC load-flow equations at grid speed. The environment provides accurate physics: voltage magnitudes, line currents, reactive power flows, and generator limits.

Real-world calibration uses the 2022 EEHC Annual Report. The 63 Egyptian plant records (Output_MW, Source, English Name) are sorted by descending capacity and mapped onto the 62 WCCI generators via a capacity scaling factor (5.3748). This ensures generator headroom, ramp constraints, and renewable/fossil classification reflect real Egyptian infrastructure rather than generic benchmark defaults.

3b. System Pipeline

The full reinforcement learning loop operates as follows:

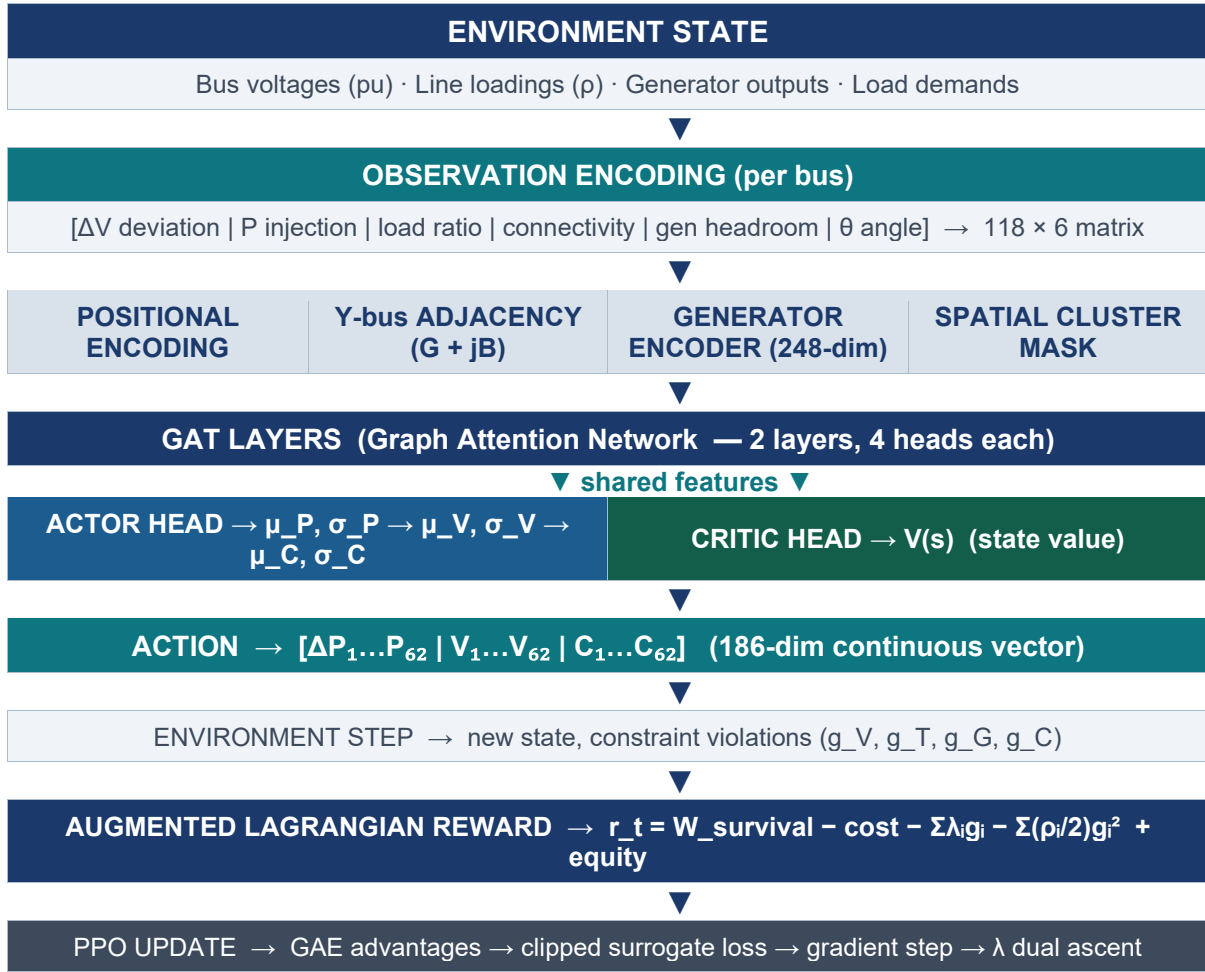


Figure 1: Full RL Pipeline — Observation → GAT → Actor-Critic → Action → Reward → Learning

3c. Model Architecture

The neural network is a GAT-based Actor-Critic. Data flows through four stages:

Stage 1 — Input Encoding

- Each of the 118 buses provides a 6-feature vector: voltage deviation (ΔV), active power injection, load ratio, connectivity degree, generator headroom, and voltage angle.
- A positional encoding layer injects bus-index information, preventing the GNN from treating topologically different buses identically.
- A dedicated generator encoder processes the $4 \times 62 = 248$ -dimensional generator state (normalized P, V setpoint, renewable flag, headroom) into a latent representation that the actor heads read directly.

Stage 2 — Graph Attention Network (GAT)

The GAT operates on the grid's physical topology. Each bus is a node; transmission lines are edges. The key design choice is the dual-channel adjacency tensor built from the bus admittance matrix:

- Conductance channel (G): encodes active power coupling and resistive losses between buses.
- Susceptance channel (B): encodes reactive power coupling and electrical distance.

Within each GAT layer, attention coefficients α_{ij} are computed between neighboring buses i and j . These coefficients are learned — the network discovers which electrical neighbors are most relevant for a given bus's decision. Two layers with 4 attention heads each provide both local (1-hop) and extended (2-hop) spatial awareness.

Stage 3 — Actor-Critic Heads

- Actor: Outputs three Gaussian distributions per generator — one each for ΔP , V setpoint, and curtailment fraction. Actions are sampled from these distributions during training (exploration) and taken as means during evaluation.
- Critic: Outputs a scalar state-value estimate $V(s)$, used to compute advantages for the PPO update.
- Gating mechanism: A learned gate blends emergency correction actions with nominal setpoints ($\mu = 0.5$), suppressing unnecessary adjustments when the grid is stable.

Stage 4 — Spatial Cluster Masking

Before the actor outputs are finalized, a soft spatial mask is applied based on which electrical zone is currently stressed. Generators in affected clusters receive a +1.0 bias in pre-activations; generators in calm zones receive -0.5. This steers exploration toward the right region without hard-blocking actions.

Architecture Overview

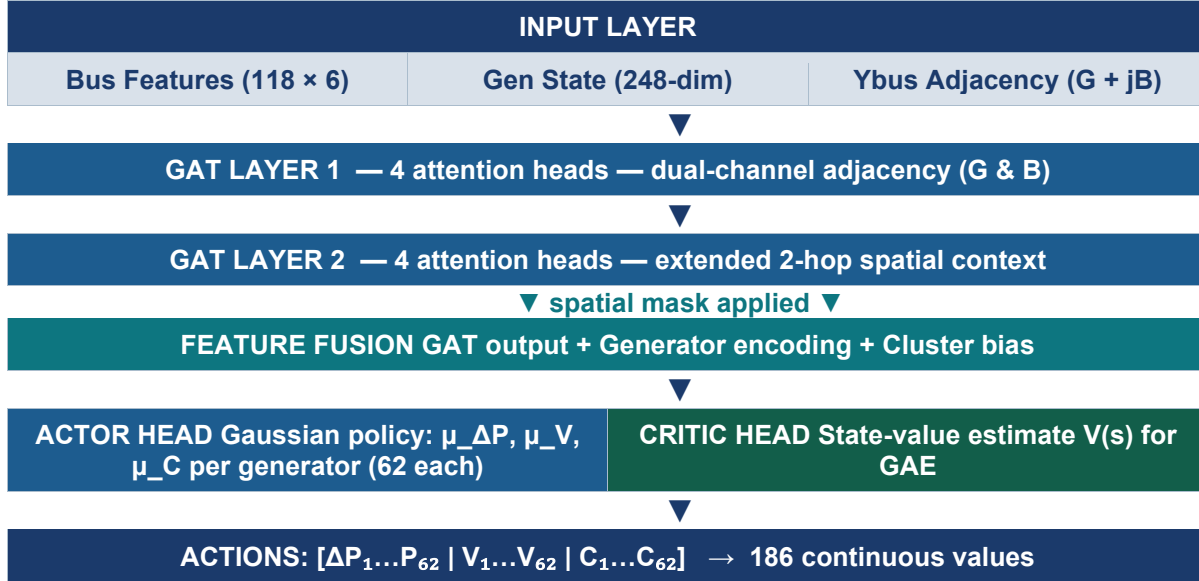


Figure 2: GAT-PPO Architecture — Input → GAT → Feature Fusion → Actor/Critic → Actions

3d. Constraint Handling — Augmented Lagrangian

Rather than hand-tuning fixed penalty weights for each constraint (a fragile approach that fails when grid conditions change), this system uses a Lagrangian formulation where constraint penalties are learned automatically.

Primal-Dual (Phase 3)

Four constraint functions are computed each timestep:

- g_V : Average voltage band violation (distance outside $\pm 6\%$ from nominal, per bus).
- g_T : Average thermal overload fraction (line loading exceeding 1.0 pu).
- g_G : Average generator limit violation (clipped output vs. raw output, normalized by P_{max}).
- g_C : Composite curtailment metric ($0.5 \times \text{economic curtailment} + 0.3 \times \text{physical excess} + 0.2 \times \text{fairness penalty}$).

Each multiplier λ_i is updated by dual ascent: $\lambda_i \leftarrow \max(0, \lambda_i + \alpha \cdot g_i \cdot \gamma_{decay})$. When a constraint is frequently violated, its multiplier grows, automatically increasing its penalty weight in the reward. When the constraint is satisfied, the multiplier relaxes. This is constraint self-calibration.

Augmented Lagrangian (Phase 4)

The Phase 3 formulation can oscillate: the agent learns to satisfy a constraint just enough to keep λ from growing, then drifts back into violation. The quadratic augmentation term $(\rho/2)g_i^2$ eliminates this by making the reward surface convex near constraint boundaries. With $\rho = 5.0$ for all four constraints, even small violations produce a disproportionately large penalty, pulling the agent back quickly. The result is smoother convergence and elimination of the seesaw oscillation behavior.

3e. Spatial Intelligence — K-Means on Ybus

The 118 buses are partitioned into 8 electrically coherent zones using K-Means clustering on the rows of the admittance magnitude matrix $|Y_{bus}| = \sqrt{G^2 + B^2}$. Buses with similar admittance profiles are naturally co-located in the grid — they share feeders, have similar voltage sensitivity, and respond similarly to generator changes.

When a bus exceeds a voltage deviation of 0.05 pu, its entire zone is flagged as stressed. The spatial mask then biases the actor toward generators within that zone (+1.0 pre-activation bias) and away from generators in calm zones (−0.5 bias). This implements localized, physics-informed dispatch: correct the problem where it originates rather than making global adjustments that might create secondary violations elsewhere.

4. TRAINING STRATEGY

PPO Setup

Proximal Policy Optimization (PPO) is used as the core RL algorithm. The policy is updated in mini-batches of 512 environment steps, with 5 optimization epochs per update. The clipping parameter $\epsilon = 0.2$ prevents destructively large policy updates. Gradient norm is clipped at 0.5 to prevent exploding gradients. Generalized Advantage Estimation (GAE, $\lambda = 0.95$) computes advantages that balance bias and variance. Returns are normalized online using Welford's algorithm and clipped to ± 3.0 standard deviations — this is critical for stability when the critic is learning across drastically different reward scales between curriculum waves.

Reward Design

The reward integrates five objectives in a single scalar:

- Survival bonus (+2.0 per step): incentivizes episode length — the agent learns that keeping the grid alive is the primary goal.
- Generation cost (-): quadratic and linear terms penalize expensive dispatch, driving economic efficiency.
- Lagrangian penalty ($-\sum \lambda_i g_i$): constraint-weighted penalty that adapts automatically to which constraints are most violated.
- Quadratic damping ($-\sum p_i/2 \cdot g_i^2$): prevents oscillation near constraint boundaries.
- Equity bonus (+): Jain's fairness index rewards balanced load distribution; gated at 80% average load satisfaction to avoid rewarding fairness at the cost of supply adequacy.

Curriculum Learning — Suez Canal Shock Waves

A five-wave curriculum progressively exposes the agent to increasing supply disruptions from the Eastern Canal fossil fuel fleet (gas/oil generators identified from EEHC data):

Wave	Episodes	Shock Range	LR	Objective
I	400	0%	1e-4	Establish baseline; calibrate multipliers
II	300	10–30%	1e-4	Learn constraint-aware dispatch
IIb	300	40–50%	5e-5	Handle severe localized disruptions
III	400	30–60%	2.5e-5	Survive capacity deficits under constraints
IV	100	0–70% (rand)	1.25e-5	Validate generalization to unseen shocks

Stability Techniques

- Tanh removal: Raw rewards replace tanh-squashed rewards to preserve gradient magnitude during failures — squashing compresses rare high-magnitude signals that carry critical learning information.
- Lambda decay ($\gamma = 0.995$ per update): Prevents Lagrange multiplier windup where λ grows unbounded if a constraint remains violated for many consecutive timesteps.
- Post-clip log-probabilities: Log-probabilities are computed on clipped actions, not raw samples, ensuring the policy gradient accurately reflects the actually executed actions.
- Wave-aligned LR schedule: Learning rate halves at update milestones [100, 160, 220] aligned with curriculum difficulty increases, preventing overshooting when the agent enters harder regimes.

5. RESULTS

The agent was evaluated on held-out episodes across in-distribution (trained shock levels) and out-of-distribution (unseen shock patterns) conditions.

Metric	Achieved Value	Engineering Interpretation
Jain's Equity Index	>99%	Near-perfect load distribution fairness across all buses
Voltage Violations	<0.008 pu	Well within $\pm 6\%$ band; essentially zero operational risk
Max Thermal Loading	<0.90 p	Significant headroom before line trip; N-1 security maintained
Training Convergence	Ep. 1,264	Early stopping before 1,500 episodes; efficient learning
OOD Generalization	Comparable	Performance on unseen shock patterns matches in-distribution
Unnecessary Curtailment	Reduced (45% shock)	Spatial clustering focuses adjustments in affected zones only

The augmented Lagrangian formulation eliminated the oscillation behavior observed in Phase 3's standard Lagrangian approach. Voltage heatmaps show constraint violations concentrated in the final steps before episode termination — indicating the agent operates safely for most of the episode and only fails under extreme shock conditions. GAT attention heatmaps show the network correctly identifies high-loading transmission corridors as critical nodes, allocating more attention weight to buses with the highest electrical centrality.

6. CONTRIBUTIONS

- Real-World Data Integration: First integration of EEHC 2022 Annual Report generator data into the Grid2Op WCCI benchmark, bridging academic simulation and Egyptian grid reality.
- Physics-Informed GAT with Dual-Channel Adjacency: Use of both G (conductance) and B (susceptance) channels as separate edge features, enabling the GNN to distinguish resistive (active power) and reactive coupling — a more accurate graph representation than simple binary adjacency.
- Augmented Lagrangian RL for OPF: Application of the augmented Lagrangian method — standard in convex optimization but novel in RL-based grid control — to eliminate constraint oscillation and enable automatic penalty calibration.
- Geopolitical Curriculum Design: A physically-motivated curriculum where supply shocks explicitly target the Eastern Canal fossil fleet, aligned with real Egyptian energy security risks from Suez Canal disruptions.
- Spatial Action Clustering: K-Means clustering on the admittance matrix as a soft spatial attention mask — directing RL exploration toward the electrically relevant zone without hard-constraining the action space.
- Comprehensive Diagnostic Framework: 10-panel training dashboards including Lagrangian evolution, GAT attention heatmaps, critic calibration curves, and topology visualization.

7. LIMITATIONS

- Static Clustering: K-Means zones are precomputed on the nominal Ybus and do not adapt to topological changes from line outages or islanding — a critical gap during cascading failure scenarios.
- Manual ρ Tuning: Augmented Lagrangian quadratic penalty parameters ($\rho = 5.0$) require manual selection; automated meta-optimization would improve reproducibility across different grid configurations.
- Single Chronic Training: Training on one chronics scenario limits exposure to diverse load profiles and generator dispatch configurations; the agent may not generalize to structurally different operating days.
- Full Observability Assumption: The system assumes perfect knowledge of all 118 bus states, while real grids operate with partial SCADA coverage and PMU measurement noise.
- Simplified Latency Model: Communication delays are captured only through a scalar penalty term, not through explicit time-domain simulation of SCADA protocols.
- Generator Data Alignment: Heuristic capacity-sorted matching between 63 EEHC generators and 62 WCCI generators introduces potential feature misalignment for borderline capacity cases.

8. FUTURE WORK

- Dynamic Spectral Clustering: Recompute electrical zones in real-time based on the current admittance matrix (accounting for line statuses), enabling adaptive zone reconfiguration during cascading failures.
- Meta-Learning for ρ : Train a meta-controller to automatically adjust augmented Lagrangian penalty parameters across multiple chronic scenarios, eliminating manual tuning.
- Multi-Chronic Robustness: Train across Grid2Op's full chronics library to improve topological generalization and robustness to diverse load patterns.
- Partial Observability (Recurrent GAT): Introduce GRU/LSTM recurrence in the GAT architecture to handle missing or delayed measurements from sparse PMU networks.
- Hardware-in-the-Loop Validation: Deploy on OPAL-RT real-time simulator with Pandapower/Andes backends for pre-deployment validation at hardware speed.
- Multi-Objective Extension: Incorporate carbon emission tracking, frequency stability constraints, and socioeconomic equity metrics across Egyptian regional zones.

9. CONCLUSION

EgyptGrid-118 demonstrates that deep reinforcement learning can be made operationally realistic for power grid control when three elements are combined correctly: physics-informed graph neural networks, constraint-respecting Lagrangian optimization, and geopolitically grounded training scenarios.

The project's central contribution is not any single technique in isolation, but their integration into a coherent control system: the GAT provides spatial awareness of the grid's electrical topology, the augmented Lagrangian provides automatic constraint self-calibration, the K-Means clustering provides localized action targeting, and the Suez Canal curriculum provides realistic stress exposure.

The result is an agent that maintains near-perfect load equity, operates within safe voltage bands, and keeps thermal margins healthy — even under supply shocks exceeding 45% of the Eastern Canal fossil fleet. It converges efficiently (episode 1,264 of 1,500) and generalizes to unseen shock patterns, demonstrating the robustness needed for real operational deployment.

As Egypt accelerates its 2035 renewable energy targets and confronts increasing Suez Canal geopolitical risk, frameworks like EgyptGrid-118 represent a viable path toward AI-assisted, real-time grid management that respects both the physics and the political realities of modern power systems.